SCIENCECLAW, MIT MEDIA LAB

Null Hypothesis

Can a collective of AI agents, each seeing one slice of the evidence and coordinating with no planner, find drug targets better than one agent that sees everything?

Team	Karuniya Premnath, Karishma Premnath, Dheepak Karan
Repository	github.com/karunprem45/natures-knockouts (public)
Submission tag	submission (commit 2307265)
Pre-registration	prereg (commit 12b8c58), set before any arm ran
Pre-registered headline	Collective vs single agent: delta AUROC +0.008 (95% CI -0.074 to +0.103), a null result
Strongest finding	Coordination penalty, paired over 5 reruns: mean -0.053, 95% CI -0.090 to -0.017 (t-test p = 0.015); negative in 5 of 5
Noise floor	0.062 AUROC, measured over 5 replicates of A3; every effect quoted against it
Discovery	ACVR1C, a panel negative with published protective loss of function (54% lower T2D risk), surfaced blind
Tags	prereg, results-raw, results-final, submission, prereg-v2

This report summarises everything built and measured during the hackathon: the idea, the system, the data, every experiment and its result, how the work was kept honest, which platforms were used, and who did what.

1. Summary

Some people are born with a broken copy of a gene and are healthier for it. Loss of **PCSK9** means lifelong low LDL cholesterol, far fewer heart attacks, and eventually a drug (evolocumab). Finding the next such gene means combining evidence that lives in separate places: rare-variant burden studies, GWAS, population tolerance data and effects on other traits.

We built a decentralised collective of specialist agents for exactly that split. Each agent can open only its own folder of evidence; they coordinate through a shared board with no central planner. We tested it against a single agent that sees everything and a strong scripted baseline, on a blind panel of 40 genes frozen in git before any experiment ran.

Result	Value
Pre-registered headline: A3 (collective) minus A1 (single agent)	**+0.008 AUROC** (95% CI -0.074 to +0.103): a tie
A3 against chance (1,000 label shuffles)	permutation p = 0.003
Coordination penalty, paired over 5 reruns (secondary)	**-0.053** (95% CI -0.090 to -0.017, p = 0.015); 5/5 negative
Measured noise floor for unpaired contrasts (5 replicates)	0.062 AUROC: most unpaired effects sit near it
Centralized planner forced to pick 2 of 4 (extension)	-0.013 vs round 0; always chose burden, never GWAS
Evo 2 on 200 coding loss-of-function variants (extension)	-0.006 vs A3: prediction not supported
Isolated specialists minus single agent (secondary)	+0.080 (-0.004 to +0.173); +0.037 on GPT-5 mini
A3 + NVIDIA Evo 2 Variant agent (extension)	+0.038 AUROC over A3 (CI includes zero)
A2 centralized planner (extension)	Planner consulted everyone, so it re-ran round 0: not a centralization test
Calibrated skeptic on a fresh panel (prereg-v2)	+0.004 (-0.032 to +0.042): no improvement
Top-ranked negative	ACVR1C: published 54% lower type 2 diabetes risk in loss-of-function carriers

Overall completion: the full BUILD.md plan (Phases 0-7) is done, plus the reusable skills, two platform extensions (NVIDIA Evo 2, OpenAI robustness) and every task in the final-gaps plan: the ACVR1C case file, the secondary comparisons, the deck rewrite, a centralized arm, a reproducibility check, and a pre-registered test of a calibrated skeptic on a fresh panel; then a measured noise floor from five replicates, a centralized planner forced to choose, and a written-down prediction for Evo 2 on coding variants. Nebius was dropped because no API key was available to our code; Stage 2 binder design was out of scope by design.

2. The system

Agent	Sees only	Question it answers
Burden	In-scope rare-variant burden rows	Do carriers get less disease?
GWAS	In-scope fine-mapped credible sets	Does a second study type agree?
Tolerance	LOEUF, pLI, DepMap essentiality	Can a human live without it?
Safety	Out-of-scope traits; Mendelian disease only as categories	Does breaking it harm elsewhere?
Skeptic	Only the other agents' verdicts (no data)	Is this evidence trustworthy?
Variant (extension)	Lead variant, scored by NVIDIA Evo 2 40B	Is the variant really disruptive?

- **Round 0:** the four specialists post independent verdicts and broadcast needs for evidence they cannot see.
- **Round 1:** each agent inspects the board and bids on the open need it can answer, ranked by the organizers' formula: pressure = 2 x novelty + centrality + 0.5 x depth + 0.2 x age.

- **Round 2:** the skeptic posts typed challenges (underpowered, single cohort, single ancestry, biomarker only, direction conflict, essential gene); agents that received answers or challenges may revise.
- **Scoring** is pure Python, never the model: the latest verdict per channel, revisions supersede parents, a challenged channel counts half.
- Every artifact carries a UUID, a content hash and parent links, so each gene has a full provenance graph. Nothing in the loop assigns work.

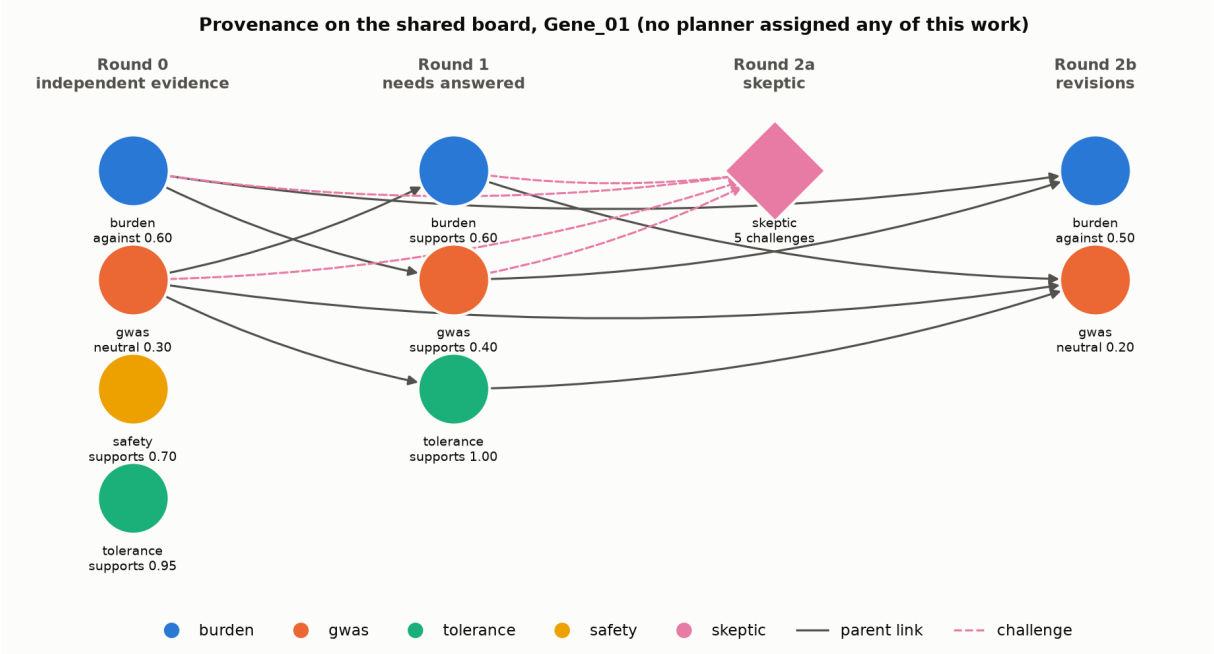


Figure 1. Provenance for Gene_01: four round-0 verdicts, three needs answered by other agents, one skeptic artifact with five challenges, and two revisions.

3. Data and the blind panel

- Source: Open Targets Platform release 26.09 (GraphQL), plus Ensembl REST for reference DNA.
- Candidate pool: 118 genes with in-scope, protective loss-of-function burden evidence (lipids, cardiovascular, metabolic, liver).
- Panel: 12 known targets (PCSK9, ANGPTL3, APOC3, ANGPTL4, NPC1L1, ASGR1, SLC30A8, MAP3K15, HSD17B13, CIDEB, GPR75, INHBE) and 28 hard negatives sampled with a fixed seed from 89 eligible genes.
- Genes are coded Gene_01 to Gene_40; labels live in a folder no agent code can read.
- **Leak found and closed:** in this Open Targets release, drug (ChEMBL) evidence is named *clinical_precedence*. It is routed only to the labels folder, and a test checks it never appears in a view.
- **Identity scrub:** 177 rows naming a gene or its own protein were dropped; Mendelian disease names were later coarsened to categories after an agent recognised a gene from eponymous syndromes.
- **Isolation test:** 203 checks prove no view holds another channel's data, a gene symbol, an Ensembl ID or drug data. A planted leak makes it fail.

4. What we built, phase by phase

Phase	What was built	Checkpoint result
0 - Setup	Repo, three-person commit credit, connection checks (Gemini, Open Targets, Ensembl, Evo 2)	Open Targets and Ensembl pass; Gemini via Vertex AI after sign-in
1 - Data	Open Targets client, pool, per-gene pulls, labels, blind panel, per-agent views, isolation test	40 genes x 5 views, 202 isolation checks pass; prereg tag set
2 - Baselines	Features, scripted arms S1-S4, metrics with bootstrap and paired delta	S4 AUROC 0.750 (0.542-0.910)
3 - Agents	Single LLM gateway with identity and label guards, call budget, logging; four agents and skeptic	25 artifacts on 5 genes, 0 errors; guard caught a real leak
4 - Board	Needs index with pressure scoring, round loop, revisions, Python scoring	10 genes: 20 cross-agent fulfilments; skeptic moved 9 scores
5 - Runs	A1 and A3, 2 seeds each, 12 zero-cost ablations, crash-safe checkpoints	0 error artifacts after a logged rerun of rate-limited genes
6 - Evaluation	Arms table, paired headline, permutation null, five figures, findings	Headline +0.008; results frozen with hashes
7 - Slides	Deck, two-minute demo replay (no API calls), demo script	Six slides incl. extensions; make demo works offline
Extensions	Multi-provider LLM layer; NVIDIA Evo 2 Variant agent; GPT-5 mini robustness run	Evo 2 +0.038; isolated specialists replicate at 0.857
Finish	Reusable skills, README, CONTRIBUTORS, three PRs, public repo, submission tag	247 tests pass; submission tag 2307265

Problems hit and how they were handled

- The hackathon GCP organisation disallows API keys, and gemini-2.5-flash was not served: switched to gemini-3-flash-preview on Vertex AI with Application Default Credentials (amendment 1).
- An agent named a gene from eponymous diseases: gene names in outputs are now redacted and flagged, and Mendelian diseases are shown as categories (amendment 2).
- Gemini sometimes wrapped JSON in a list, wasting 18% of calls on re-asks: the parser now unwraps it.
- Running four arms at once hit rate limits: 16 gene-runs were rerun under one rule for every arm, logged.
- A race condition in the Variant agent mislabelled one flag across parallel genes: fixed, with a regression test.

5. Results (pre-registered)

40 genes, 12 positives (chance AUPRC 0.30). LLM arms are averaged over 2 seeds. Confidence intervals come from 1,000 gene resamples; deltas are paired (both arms scored on the same resample).

Arm	What it is	AUROC (95% CI)	AUPRC	Calls / gene
S1	LOEUF alone	0.720 (0.559-0.868)	0.435	0
S2	Burden strength alone	0.662 (0.487-0.827)	0.396	0
S3	GWAS strength alone	0.440 (0.270-0.587)	0.276	0
S4	All features, z-scored, equal weights	0.750 (0.542-0.910)	0.697	0
A1	Single agent, all views	0.774 (0.616-0.903)	0.653	3
A3	Collective on the board	0.782 (0.641-0.895)	0.630	10
A3 round 0	Isolated specialists (re-scored)	0.853 (0.729-0.948)	0.736	0 extra
A3 no skeptic	Challenges removed (re-scored)	0.844 (0.713-0.939)	0.727	0 extra
A3 no burden	Burden agent removed (re-scored)	0.732 (0.589-0.862)	0.545	0 extra

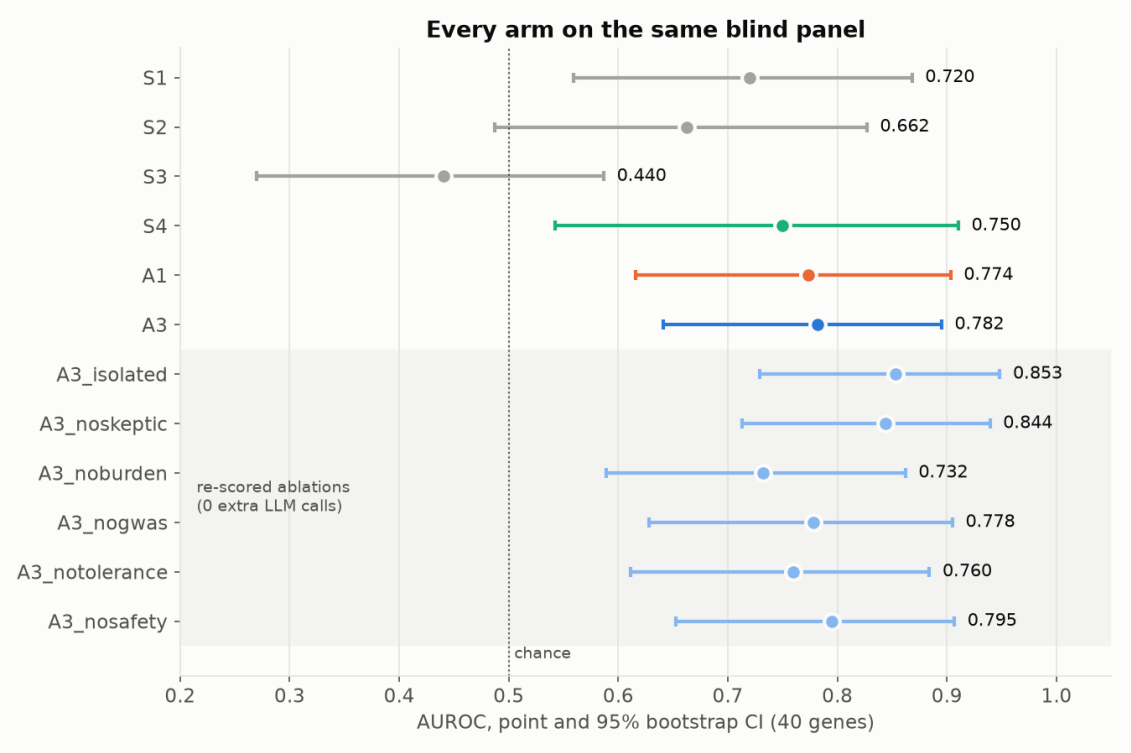


Figure 2. Every arm on the same blind panel, with 95% bootstrap intervals. Shaded rows are ablations re-scored from saved boards.

Why the coordination rounds hurt

The skeptic challenged the burden channel on 7 of 12 known targets and 21 of 28 negatives (seed 1), about the same rate whatever the truth, mostly as *biomarker only*. Halving the strongest channel almost everywhere pushed the known targets' mean rank from 10.7 to 12.5. Revisions barely moved verdicts. The headline stays A3 vs A1 as pre-registered; the ablations are reported as secondary findings and nothing was re-tuned after seeing them.

6. Extensions (after results-final, not pre-registered)

Run	Model(s)	Seeds	AUROC (95% CI)	Paired delta
A3 + Variant	Gemini 3 Flash + NVIDIA Evo 2 40B	2	0.820 (0.671-0.929)	+0.038 vs A3
A1	gpt-5-mini	1	0.820 (0.674-0.940)	-
A3	gpt-5-mini	1	0.751 (0.574-0.902)	-0.068 vs A1
A3 round 0	gpt-5-mini	1	0.857 (0.730-0.961)	+0.037 vs A1

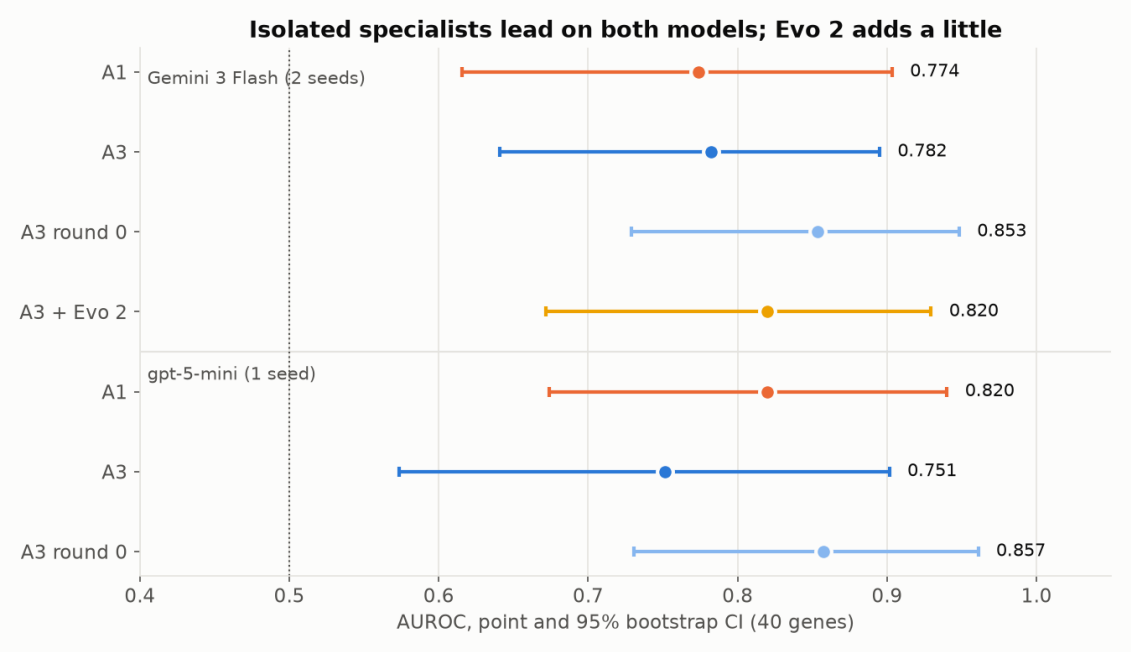


Figure 3. The same arms on two models, plus the Evo 2 Variant agent.

- **NVIDIA Evo 2:** one forward pass over 2,048 bp around the lead variant scores every possible base change in the window; the agent reports the variant's percentile and its log-likelihood change. Agents asked for it 77 times; Evo 2 answered 30 with no errors. Coordinates never reach a prompt.
- Most GWAS lead variants are non-coding tags (median percentile 59), so Evo 2 had little disruption signal to add. A fair test would feed it coding or burden-test variants on a fresh, pre-registered panel.
- **GPT-5 mini** rejects temperature 0, so it ran at its default temperature.
- **What replicates:** isolated specialists score about 0.85 on both models, and the full loop scores lower than its own round 0 on both.

7. Final gaps: post-freeze work (not pre-registered)

Everything in this section was added after results-final. The frozen chain was never touched: all 70 files under frozen/ and frozen_results/ were checked byte-identical at the start and end, and every SHA-256 verifies. Outputs live under exploratory/.

7.1 Secondary comparisons from frozen scores

Rubric comparison	Model	Contrast	Paired delta AUROC (95% CI)
Collaborative vs isolated	Gemini (2 seeds)	A3 - round 0	**-0.071 (-0.139 to -0.013)**, excludes 0
Collaborative vs isolated	GPT-5 mini (1 seed)	A3 - round 0	-0.106 (-0.239 to +0.008)
Multi-agent vs single-agent	Gemini	round 0 - A1	+0.080 (-0.004 to +0.173)
Multi-agent vs single-agent	GPT-5 mini	round 0 - A1	+0.037 (-0.084 to +0.158)
Pre-registered headline	Gemini	A3 - A1	+0.008 (-0.074 to +0.103)

The round-0 arm is a pre-registered ablation, but choosing these contrasts to highlight came after seeing results, with no multiplicity correction. Precision@12 and cost per arm were already in the frozen arms table (P@12 is pre-registered).

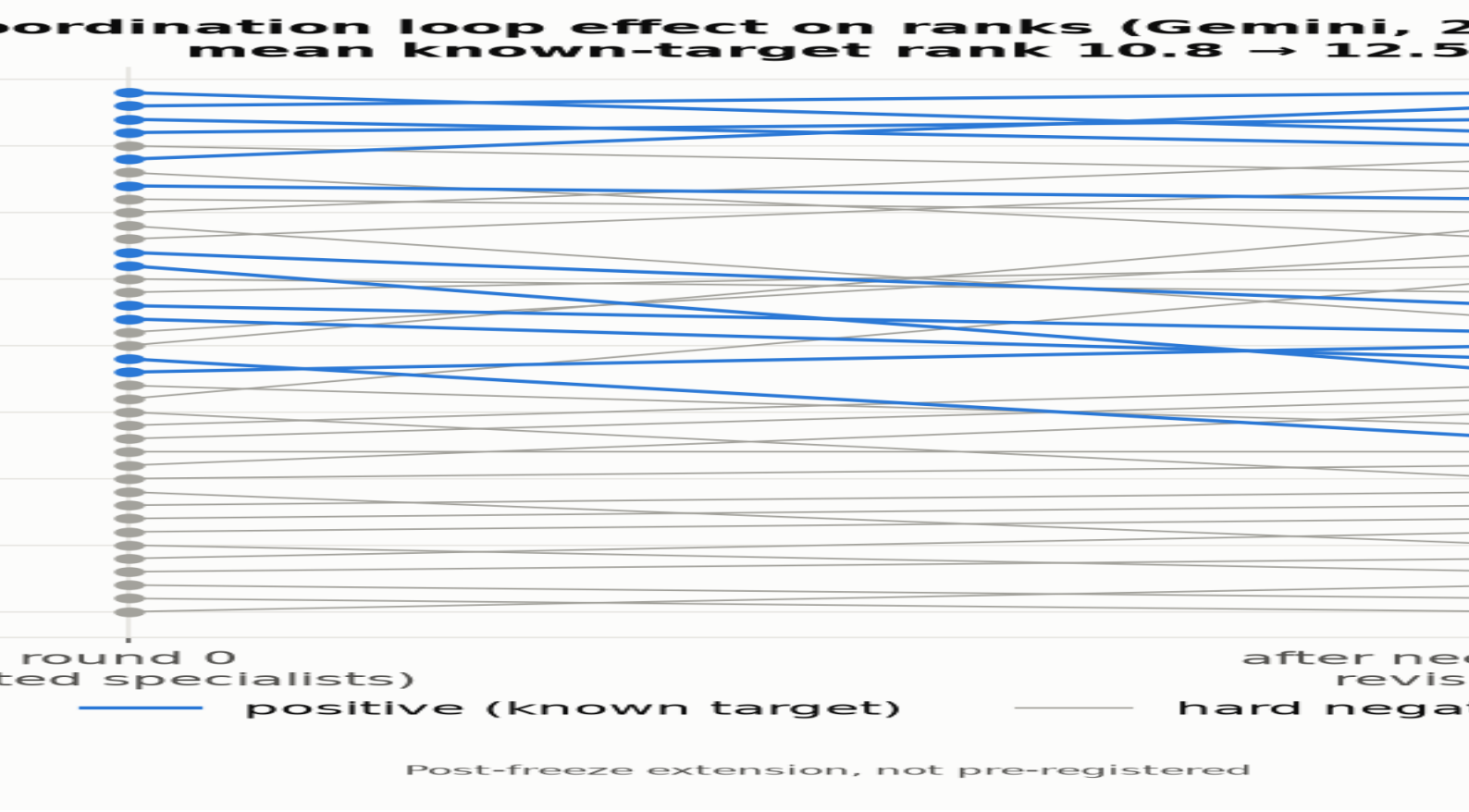


Figure 4. Per-gene rank from round 0 to the final A3 score (Gemini, 2 seeds). Known targets move from mean rank 10.8 to 12.5.

7.2 ACVR1C case file

A3 run	Rank of 40	Rank of 28 negatives
Seed-averaged	6	1
Seed 1	4	2
Seed 2	15	8

- Blind: 0 identity terms (symbol, Ensembl ID, 12 synonyms) and 0 drug terms in its 5 view files; its 5 isolation tests pass; no output flags.

- Seed 2's drop is the skeptic: it fired biomarker_only on burden and GWAS and halved both channels.
- Published evidence (Emdin et al., *Diabetes* 2019;68:226-234): carriers of predicted damaging ACVR1C variants had 54% lower type 2 diabetes risk (OR 0.46, 95% CI 0.27-0.81, exome sequencing of 55,516 people); in 405,569 UK Biobank participants, lower waist-to-hip ratio through ACVR1C meant 30% lower risk (OR 0.70, P = 5.6 x 10^{-13}).
- Framed as surfaced blind, not discovered: the 2019 evidence predates the run. ZNF229 and PKD1L3 stay hypotheses.

7.3 A2: centralized planner (1 seed)

Everything held constant except who decides the trajectory. A planner LLM saw only the gene code and one line per specialist, never evidence. It consulted all four specialists for all 40 genes, so A2 re-ran the round-0 configuration: 23 of 40 genes scored identically. A2 AUROC 0.871; A2 - round 0 = +0.028 (-0.008 to +0.090) is the gap between two runs of one configuration, not a centralization effect (the floor measured properly over five replicates is 0.062; see 7.6). Cost $0.60.

7.4 Reproducibility

- Clean clone: tests pass (8 data-dependent tests skip with a reason until the evidence is rebuilt); make demo runs with no keys and no network.
- verify_frozen: 63/63 hashes match, all 12 arms reproduce their AUROC from frozen scores, and round 0 re-scored from frozen boards matches exactly. Labels published in reproduce/ after results-final.
- REPRODUCE.md lists which steps run offline, which need network, and which need keys.

7.5 A calibrated skeptic on a fresh panel (prereg-v2)

Per-challenge-type discounts were fitted on panel 1 with one formula: multiplier = 1 - 0.5 x max(0, (rate on negatives - rate on positives) / rate on negatives). They were committed and tagged prereg-v2 before panel 2 existed.

Challenge type	Rate on positives	Rate on negatives	Multiplier
biomarker_only	0.83	0.84	1.00 (ignored)
direction_conflict	0.00	0.29	0.50
essential_gene	0.00	0.21	0.50
single_ancestry	0.25	0.48	0.76
single_cohort	0.08	0.36	0.62
underpowered	0.04	0.09	0.73

Panel 2 (12 shared positives, 28 fresh negatives)	AUROC (95% CI)
Calibrated skeptic	0.932 (0.835-0.997)
Flat 0.5 skeptic	0.927 (0.828-0.991)
Round 0 (isolated specialists)	0.954 (0.885-0.997)
Primary: calibrated - flat	**+0.004 (-0.032 to +0.042)**

Outcome, in the pre-registered wording: no improvement. The effect is far inside the noise floor (0.03 from one pair of runs at the time; 0.062 when measured over five replicates, 7.6). Panel 2 is easier (round 0 at 0.954), so the flat skeptic cost only -0.027 and there was little to recover. The pre-registration set no threshold for improvement; the conservative rule (must beat the noise floor and exclude zero) was decided when results were read, and the report says so. 203 isolation checks passed on panel 2; one A3 run, 400 calls, 0 errors, $1.38.

7.6 The noise floor, measured

A3 was re-run with seeds 3, 4 and 5 under identical settings (0 errors) and combined with the frozen seeds 1 and 2. The earlier 0.03 floor came from one accidental pair of runs; measured over 10 pairs it is about twice that.

Measure	Value
AUROC per replicate (seeds 1-5)	0.790, 0.774, 0.829, 0.842, 0.787
Pairwise |delta AUROC|, n = 10	mean 0.036, sd 0.023, median 0.040, 95th percentile 0.062
Measured noise floor	**0.062** (95th percentile; max pairwise 0.068)
Coordination penalty per replicate	-0.052, -0.091, -0.040, -0.013, -0.070 (mean -0.053)
Sign of the penalty	negative in 5 of 5; one-sided sign test p = 1/32 = 0.031

Effect	Size	x floor
Coordination penalty, frozen seeds	0.071	1.1x
Coordination penalty, mean of 5 replicates	0.053	0.9x
Isolated specialists vs single agent	0.080	1.3x
Evo 2 Variant channel (lead tags)	0.038	0.6x
Constrained planner vs round 0	0.013	0.2x
Evo 2 on coding variants	0.006	0.1x
Calibrated skeptic, panel 2	0.004	0.1x

This weakens the unpaired claims: they sit near the floor. The coordination penalty is different: it re-scores one board two ways, so rerun noise is largely shared by both sides and the right test is paired, across replicates. (Comparator chosen after seeing results; see 7.1.)

Paired analysis of the penalty (5 replicates)	Value
Mean, SD, standard error	-0.053, 0.029, 0.013
95% t-interval (df = 4)	**-0.090 to -0.017**
Paired t-test, two-sided	t = -4.06, p = 0.015
Bootstrap over replicates (10,000 resamples)	-0.075 to -0.030
Sign test	5/5 negative, one-sided p = 0.031

All three exclude zero. With n = 5 the t-interval rests on unverifiable normality and the bootstrap draws from only 126 distinct resamples, so none is precise; the sign test needs no assumption. The 0.062 floor remains the yardstick for unpaired contrasts such as round 0 vs A1. Details: exploratory/noise_floor/PAIRED_ANALYSIS.md.

7.7 A centralized planner forced to choose

Same as A2, except the planner must pick exactly 2 of the 4 specialists, still without seeing evidence; the two run exactly as in A3 round 0. AUROC 0.829 (seed 1). Versus round 0: -0.013 (-0.112 to +0.093), 0.2x the floor; versus A3: +0.039; versus A1: +0.039. The planner chose burden for all 40 genes and never chose GWAS (burden + tolerance 25, burden + safety 15). Two specialists that include the dominant burden channel capture most of the signal. Because the planner sees no evidence, its second pick cannot reflect the gene, so this tests a mostly fixed subset more than adaptive planning. Cost $0.32.

7.8 Evo 2 on coding variants (prediction written first)

Prediction, committed before any variant was collected: coding loss-of-function variants would lift the Variant channel above +0.038. 200 predicted loss-of-function variants from Ensembl (5 per gene; 188 stop-gained), scored on NVIDIA's hosted Evo 2 40B with a 4,096 bp window and a reference-base check: 200 scored, 0 errors.

Input to the Variant agent	Median site percentile	Paired delta vs A3 (seed 1)
GWAS lead tags (earlier)	59	+0.007 (+0.038 over 2 seeds)
Coding loss-of-function variants	4.4	**-0.006 (-0.165 to +0.132)**

Verdict: not supported, and 0.1x the floor. The coding variants were far more disruptive to Evo 2, but about equally so in known targets (median 3.4) and negatives (5.2). Evo 2 shows that a variant breaks a gene, not that breaking it helps the carrier, and every panel gene was chosen for protective loss-of-function evidence. Better variants were necessary but not sufficient. Cost $1.42; Evo 2 free.

The lesson. We wrote a prediction down, tested it, it failed, and we learned what the model can and cannot tell us. A fair test of the Variant channel needs a panel where loss of function is harmful for some genes and protective for others, so that disruption has something to separate. See exploratory/evo2_coding/WHAT_WE_LEARNED.md.

8. How the work was kept honest

Safeguard	Detail
Pre-registration	prereg tag (12b8c58): panel, metrics and scoring weights frozen before any arm ran
Raw results frozen	results-raw tag (28a0e95): scores and boards hashed before any A1/A3 metric
Final results frozen	results-final tag (4729a6b); SHA-256 hashes for every output file
Blindness	Coded genes, per-agent folders, prompt identity guard, output redaction, 203-check isolation test
Fair comparison	Same model, same 10-call cap per gene, same scoring for A1 and A3; similar spend ($1.09 vs $1.32 per run)
Amendments	Two, both before any LLM arm ran: model change, and Mendelian diseases as categories
Reruns	Only API-error genes, one rule for every arm, logged in RERUN_LOG.md
Second pre-registration	prereg-v2 tagged before panel 2 existed; frozen chain verified untouched all night

Limitations, stated up front

- 40 genes: every interval is wide; S4, A1 and A3 overlap.
- Most effects sit near the measured noise floor (0.062); only the coordination penalty's sign replicates cleanly (5 of 5 reruns).
- The secondary contrasts were chosen after seeing results, with no multiplicity correction.
- The centralized planner chooses without evidence by design; even forced to pick 2 of 4 it tests a mostly fixed subset more than adaptive planning.
- ACVR1C is the top negative only on the seed-averaged score (4th in seed 1, 15th in seed 2).
- Famous targets carry far more evidence rows, so any arm can partly score how well studied a gene is.
- Names, IDs and drug data are removed, but trait patterns remain (2 of 800 A3 artifacts still named a gene).
- Open Targets labels quantitative traits 'protective' from the sign of beta, so lower HDL reads as protective.
- Top-ranked negatives (ACVR1C, ZNF229, PKD1L3) are hypotheses, not validated targets.

9. Platforms used

Platform	Used for	Status
Google Cloud, Vertex AI (Gemini 3 Flash)	Every agent in every pre-registered arm	Used
NVIDIA build.nvidia.com (Evo 2 40B)	Variant agent	Used (free trial endpoint)
OpenAI (gpt-5-mini)	Robustness check on a second model	Used (hackathon credit)
Open Targets, Ensembl	Genetics evidence and reference DNA	Used
GitHub	Repository, tags, PRs, collaborators	Used (public)
NVIDIA Brev	Fallback GPU host for Evo 2	Not needed: hosted Evo 2 worked
Nebius	Planned open-model arm	Dropped: no API key usable by our code

Approximate spend: about $16.20 of Gemini (4,624 calls, including every extension), about $1.20 of OpenAI credit, and nothing for Evo 2 (about 450 hosted calls on the free trial endpoint). Brev's $150 was not used.

10. Team and contributions

Person	Role	Owned	Commits
Karuniya Premnath	Data	phase-0-probes, phase-1-data, fix-json-list, phase-7-slides, PR #1 (deck), deck rewrite and every slide update	13
Karishma Premnath	Agents	phase-3-agents, phase-3b-safety-coarsen, phase-4-board, explore-platforms (Evo 2), PR #2 (skills), A2 arm, reproducibility, panel-2 run, constrained planner, Evo 2 prediction, this report	13
Dheepak Karan	Evaluation	phase-2-baselines, phase-5-arms, phase-6-eval, explore-openai, PR #3 (README), ACVR1C case, secondary comparisons, prereg-v2, noise floor, Evo 2 coding run	13

We worked as a mob on one machine; every commit lists the other two as co-authors. Authorship was reassigned by phase owner after the event; file contents are identical, the prereg hash is unchanged, and the original history is kept on archive/original-history.

11. Deliverables

Deliverable	Where
Code and tests (247 passing)	github.com/karunprem45/natures-knockouts
Pre-registered results	frozen_results/arms_table.md, FINDINGS.md (SHA-256 verified)
Post-freeze work	exploratory/: ACVR1C_CASE.md, explore_table.md, A2/, v2/, noise_floor/, A2_constrained/, evo2_coding/
Six-slide deck	slides/deck.html (also published as a private artifact; share before judging)
Two-minute demo	make demo GENE=Gene_01 (replays a frozen board, no API calls); DEMO.md
Reproduction	REPRODUCE.md; exploratory.scripts.verify_frozen; reproduce/ labels
Reusable skills	skills/: blind-agent-views, needs-board-coordination, evo2-variant-score
Documentation	README.md, CONTRIBUTORS.md, BUILD.md, frozen/AMENDMENTS.md

Still to do before judging

- Share the deck from its Share menu.
- Rehearse the demo once, out loud.
- Discord, GitHub invites and the Infinite post (handled by a team member).
- Teammates: accept the GitHub invites and re-clone.

- After the event: revoke the NVIDIA and OpenAI keys that were pasted in chat.